SmartEnv - IoT-Based Smart Water & Air Quality Monitoring System

Team Name: Code-x

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Member: Anand Maurya (+919935083474)

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1. Problem Statement

India faces a severe environmental crisis with both water and air pollution reaching dangerous levels. According to the Central Pollution Control Board (CPCB), over 351 river stretches are polluted, and major cities regularly exceed WHO safe limits for air pollutants by 2-3 times.

Current Challenges

- **Manual Testing:** Lab testing takes 24-72 hours, delaying response
- **High Cost:** Professional testing costs ₹2,000-5,000 per sample
- **Limited Coverage:** Rural areas lack any monitoring infrastructure
- **No Real-time Alerts:** Communities are unaware of pollution spikes

Impact

- 37.7 million Indians affected by waterborne diseases annually
- Pollution costs India 2.9% of GDP (~$560 billion)
- Vulnerable communities bear disproportionate burden

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2. Progress Made

2.1 Hardware Development

Component	Status	Details
ESP32 Setup	Completed	WiFi + BLE configured, tested with basic sensors
Air Quality Sensors	Completed	MQ135 (CO_2, NH_3, NO_x) calibrated and working
Water Quality Sensors	In Progress	pH and TDS sensors ordered, awaiting delivery
LoRa Module	Planned	For long-range rural deployment

2.2 Software Development

Component	Status	Details
Firebase Backend	Completed	Real-time database configured, data ingestion working
ESP32 Firmware	Completed	Sensor data collection and WiFi transmission working
React Dashboard	In Progress	UI framework set up, real-time data display partial
Alert System	Planned	SMS/Email integration pending

2.3 Milestones Achieved

- [x] ESP32 reads air quality data (PM2.5, CO2)
- [x] Data transmits to Firebase in real-time
- [x] Firebase stores time-series data
- [x] Basic React dashboard displays live readings
- [x] GitHub repository set up with clean code structure
- [ ] Water quality sensor integration
- [ ] Threshold-based alert system
- [ ] Historical data visualization
- [ ] Mobile responsive design

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3. Features Implemented

3.1 Completed Features

Feature	Description
Real-time Data Collection	ESP32 continuously reads sensor values every 30 seconds
Cloud Data Storage	Firebase Realtime Database stores all sensor readings
Basic Dashboard	Web interface shows current air quality readings
Data Logging	All readings stored with timestamps for historical analysis

3.2 In-Progress Features

Feature	Description	Progress
Interactive Charts	Line graphs for historical trends	60% complete
Color-coded Alerts	Visual indicators for safe/warning/danger levels	40% complete
Multi-station Support	Monitor multiple locations from one dashboard	30% complete

3.3 Demo Screenshots

(Add screenshots here showing:)

1. ESP32 serial monitor showing sensor readings
2. Firebase console showing real-time data
3. React dashboard with live data display

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4. Technical Architecture

4.1 System Overview

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┌─────────────────────────────────────────────────────────┐
│                      SENSOR LAYER                        │
│  ┌──────────┐  ┌──────────┐  ┌──────────┐               │
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│  │ MQ135   │  │  pH     │  │  TDS    │                   │
│  │(Air)    │  │(Water)  │  │(Water)  │                   │
│  └────┬────┘  └────┬────┘  └────┬────┘                   │
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        │               │               │
        ▼               ▼               ▼
┌─────────────────────────────────────────────────────────┐
│                 ESP32 MICROCONTROLLER                     │
│         (Data Collection + WiFi Transmission)             │
└─────────────────────────┬───────────────────────────────┘
                          │ HTTP/MQTT
                          ▼
┌─────────────────────────────────────────────────────────┐
│                    FIREBASE CLOUD                         │
│  ┌─────────┐  ┌─────────┐  ┌─────────┐                   │
│  │ Realtime │  │ Auth    │  │ Hosting │                   │
│  │ Database │  │         │  │         │                   │
│  └─────────┘  └─────────┘  └─────────┘                   │
└─────────────────────────┬───────────────────────────────┘
                          │
                          ▼
┌─────────────────────────────────────────────────────────┐
│                  REACT DASHBOARD                          │
│  ┌─────────┐  ┌─────────┐  ┌─────────┐                   │
│  │  Live   │  │Historical│  │  Alert  │                   │
│  │  Data   │  │  Charts │  │  Panel  │                   │
│  └─────────┘  └─────────┘  └─────────┘                   │
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4.2 Tech Stack

Layer	Technology
Hardware	ESP32, MQ135, pH Sensor, TDS Sensor
Firmware	Arduino C++
Backend	Firebase Realtime Database
Frontend	React.js, Tailwind CSS, Chart.js
Hosting	Firebase Hosting

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5. Challenges Faced

5.1 Technical Challenges

Challenge	Impact	Solution
Sensor Calibration	Initial readings inaccurate	Researched calibration procedures, implemented offset corrections

WiFi Stability	ESP32 disconnects intermittently	Added auto-reconnect logic, implemented data buffering
Firebase Latency	Occasional delayed updates	Optimized database structure, reduced payload size
Power Supply	Sensors need stable voltage	Used separate 5V supply for sensors, added capacitors

5.2 Resource Constraints

Constraint	Mitigation
Limited budget (₹6,500)	Used cost-effective sensors, prioritized essential components
2-person team	Divided responsibilities: Anirudh (Hardware), Anand (Software)
Time pressure (3 days)	Focused on MVP first, planned enhancements for Phase 3

5.3 Learning Curve

- Firebase real-time database queries
- ESP32 WiFi management and deep sleep modes
- Sensor data normalization and calibration
- React hooks for real-time data updates

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6. Future Roadmap

6.1 Phase 3 (Final Submission - 1 September)

Task	Priority	Owner
Complete water quality sensor integration	High	Anirudh
Implement threshold-based SMS/Email alerts	High	Both
Add historical data visualization	High	Anand
Mobile responsive design	Medium	Anand
Documentation and README	Medium	Both
Demo video recording	Medium	Both

6.2 Post-Hackathon Enhancements

Enhancement	Timeline	Impact
AI-based pollution prediction	2-4 weeks	Proactive alerts
Mobile app (Flutter)	1-2 months	Better accessibility
LoRa integration	1 month	Rural deployment
Solar power option	2 months	Off-grid capability
Multi-language support	2 weeks	Wider reach

6.3 Long-term Vision

- Integration with government CPCB monitoring network
- Citizen science platform for crowdsourced data
- Blockchain-based data integrity for regulatory compliance
- Drone-based sensor deployment for remote areas

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7. Team & Responsibilities

Role	Member	Responsibilities
Hardware Lead	Anirudh Yadav	Sensor setup, ESP32 firmware, circuit design
Software Lead	Anand Maurya	Firebase backend, React dashboard, documentation

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8. Conclusion

SmartEnv addresses a critical environmental monitoring gap in India with an affordable, scalable IoT solution. Despite challenges with sensor calibration and resource constraints, we have successfully:

- Built a working prototype with air quality monitoring
- Established real-time cloud data storage
- Created a functional web dashboard

With continued development, SmartEnv has the potential to empower communities with real-time environmental data, enabling faster response to pollution events and contributing to a healthier India.

Team Code-x

GitHub: https://github.com/mauryaanand416/SmartEnv-IoT-Monitor

Submission Date: 27 August 2026

Deadline: 27 August 2026, 11:59 PM IST